

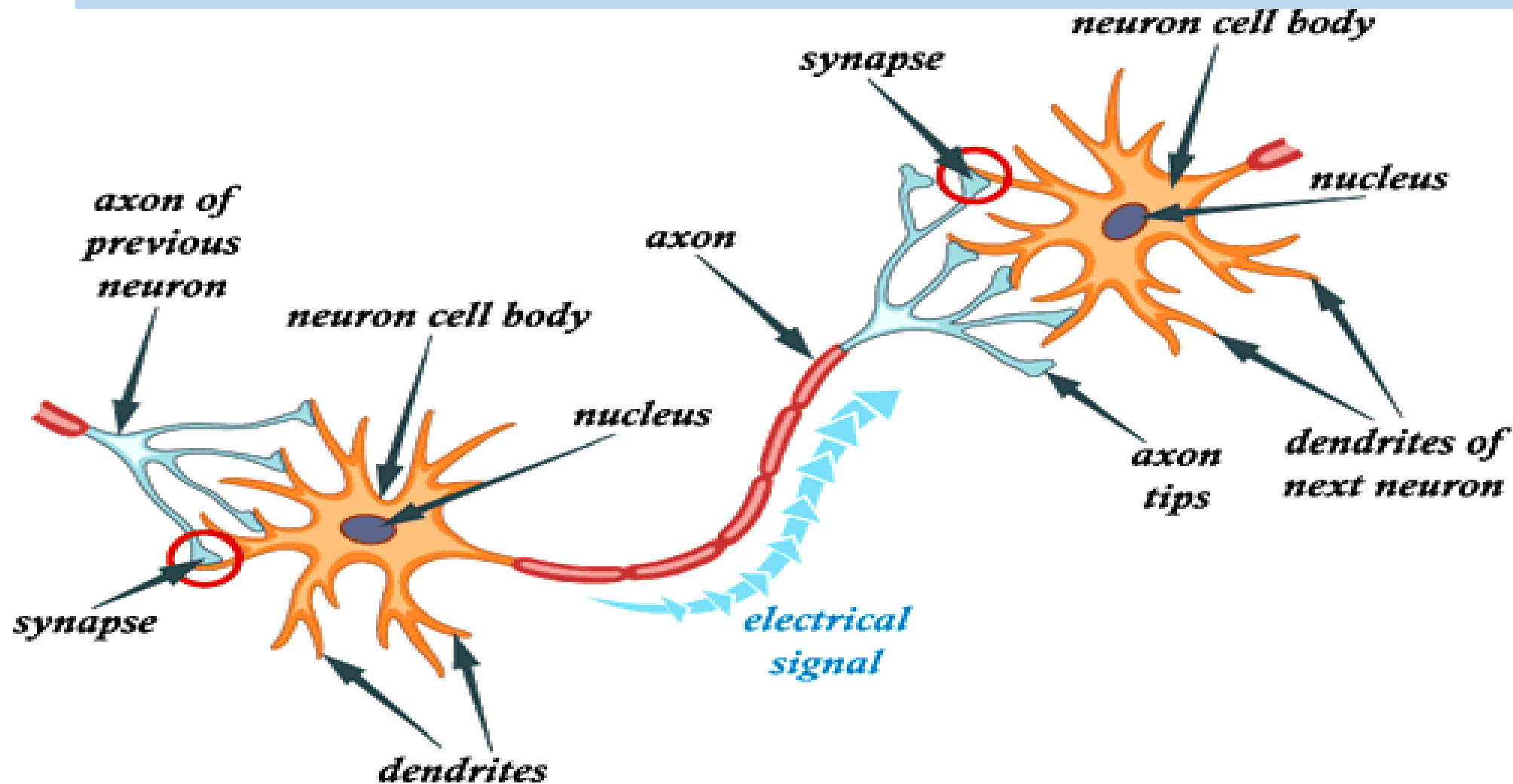
**Anatomy of
Nervous system(NS)**

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Nervous system (NS) is formed of

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- **1. Central nervous system (CNS) :** includes the **brain & spinal cord.**
- **2. Peripheral nervous system (PNS) :** supplies somatic structures
 - - **12 pairs of cranial nerves and the sensory ganglia related.**
 - - **31 pairs of spinal nerves** and the dorsal root ganglia.
- **3. Autonomic nervous system (ANS) :** is formed of the **sympathetic and parasympathetic systems.** The ANS supplies the smooth muscles, heart and glands.
-

The neuron = Structural unit = cell body + axons



1-Central Nervous system(CNS)

Central Nervous System

1. The Brain is protected by the skull
2. The spinal cord is protected by vertebral column

• The Brain: consists of

1. Cerebrum:

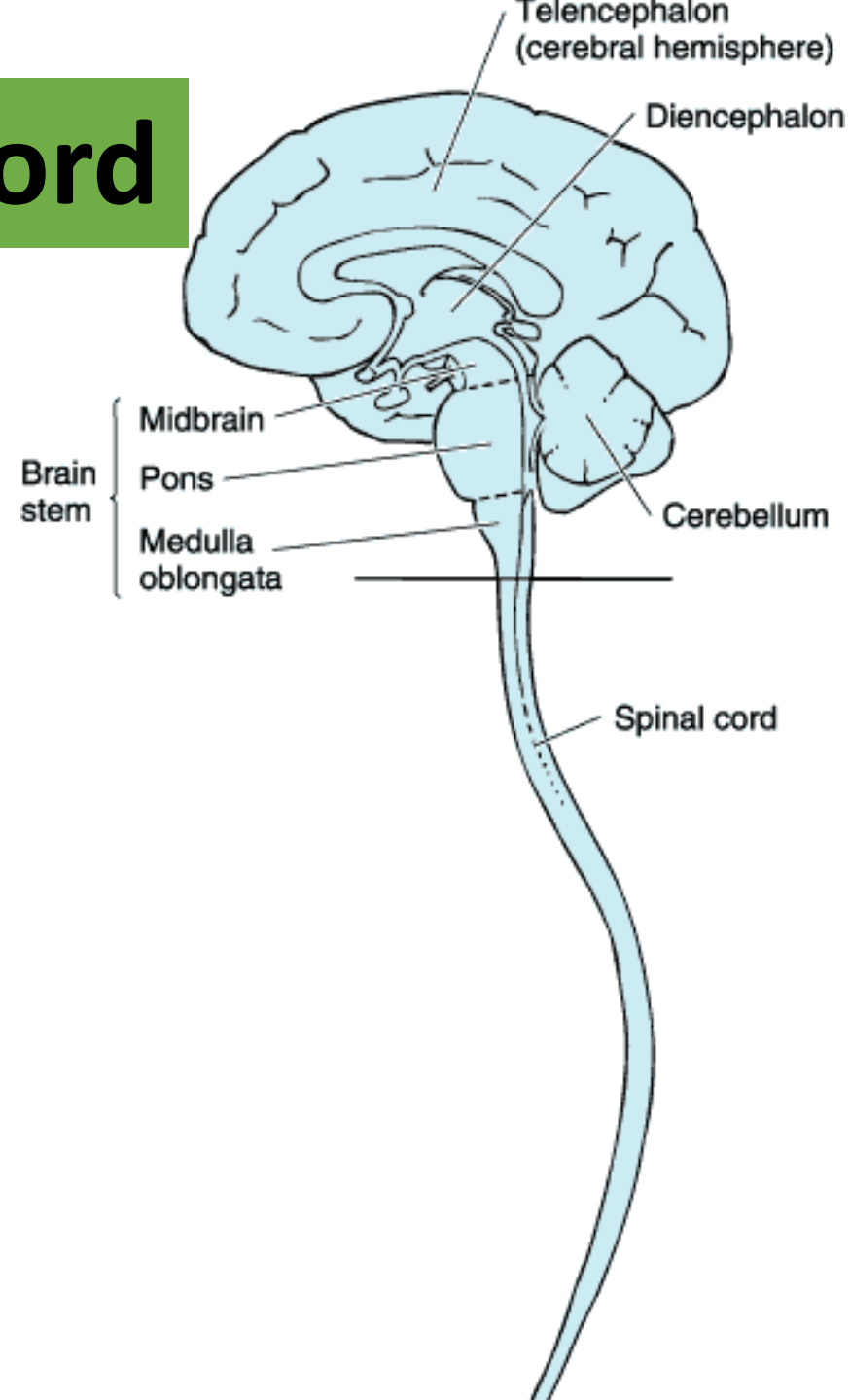
A. 2 cerebral hemispheres

B. Diencephalon which includes thalamus & hypothalamus

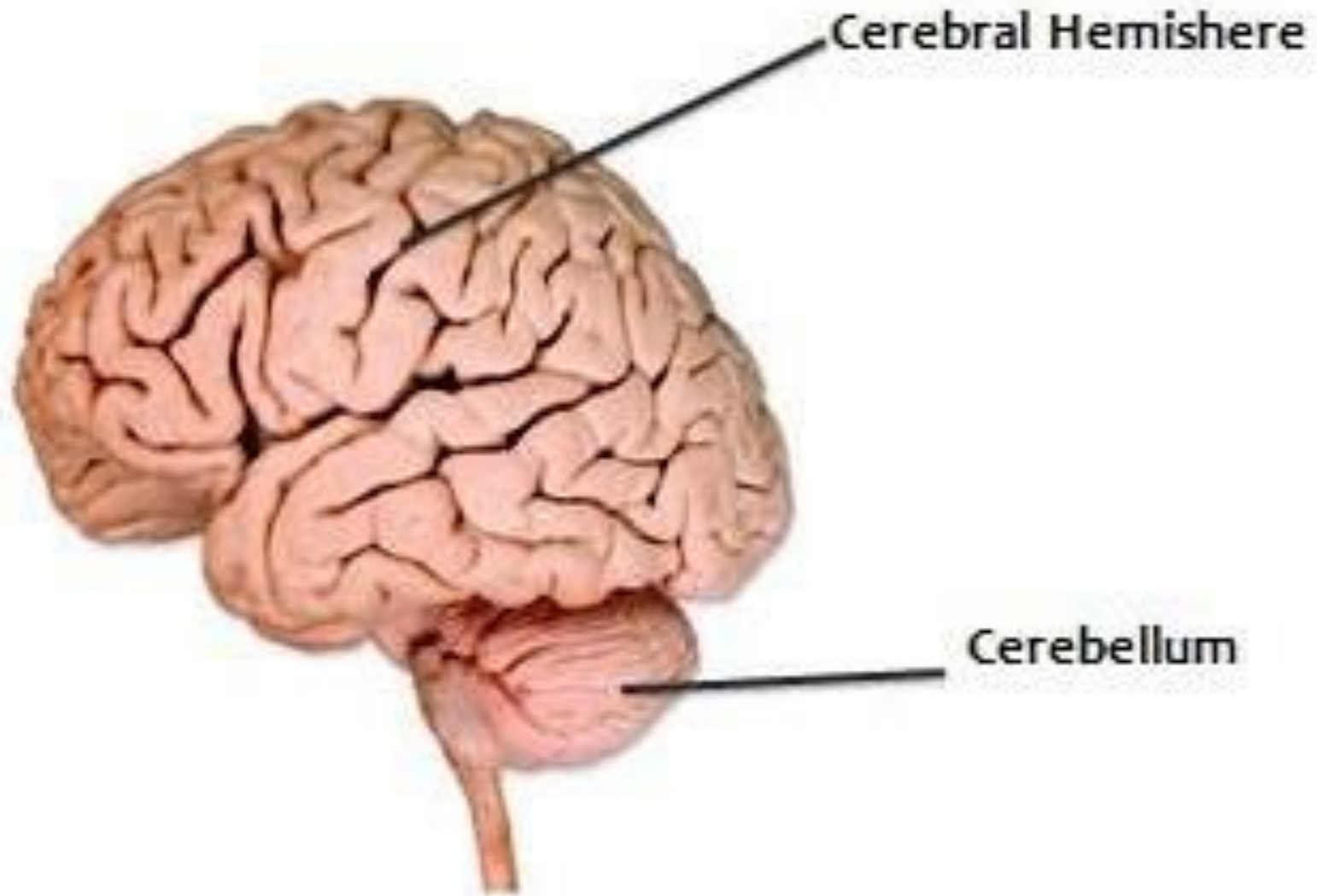
2 Brain stem: midbrain, pons & medulla oblongata

3 Cerebellum: consists of 2 cerebellar hemispheres

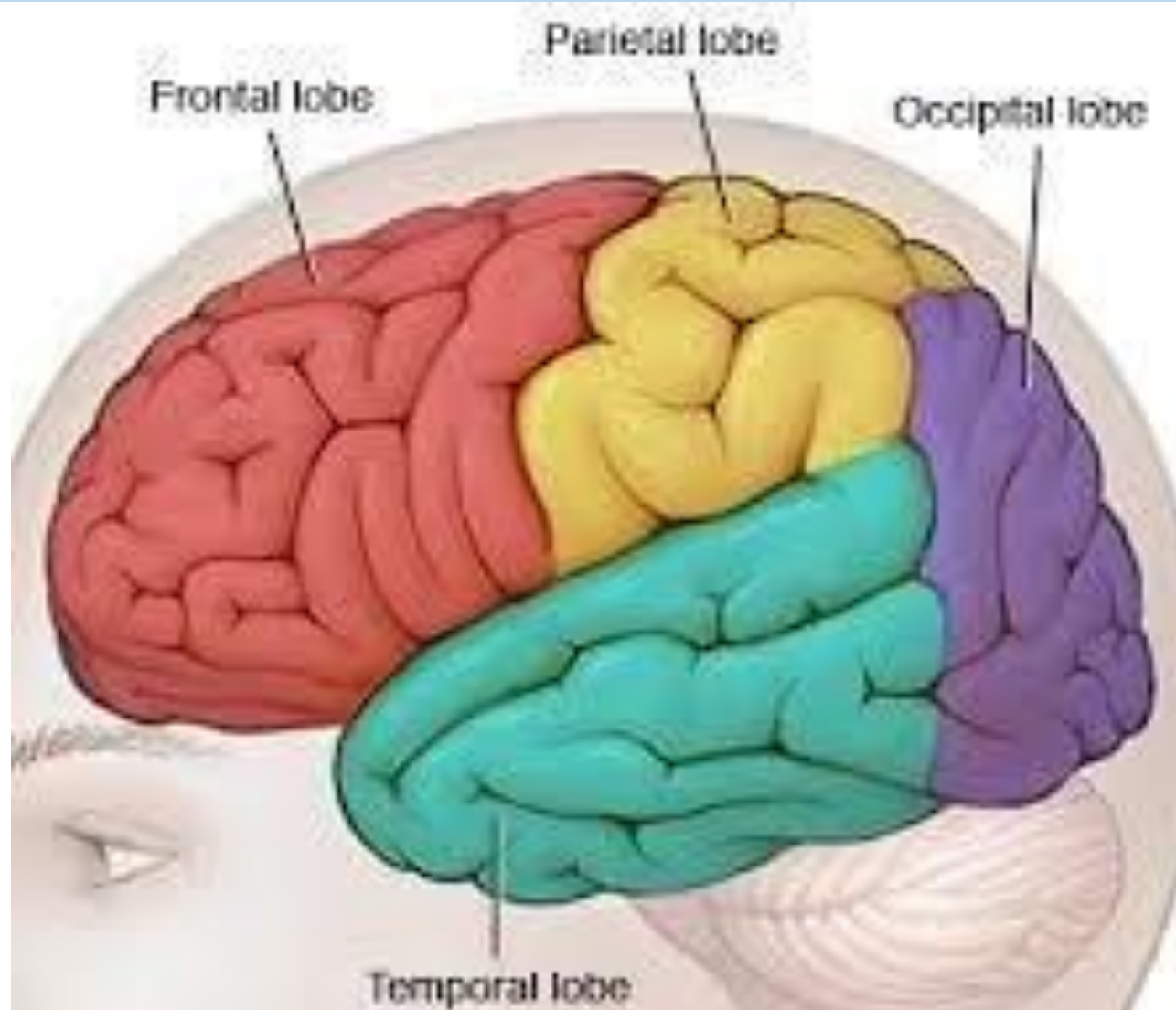
Brain & spinal cord



Cerebral hemisphere& Cerebellum

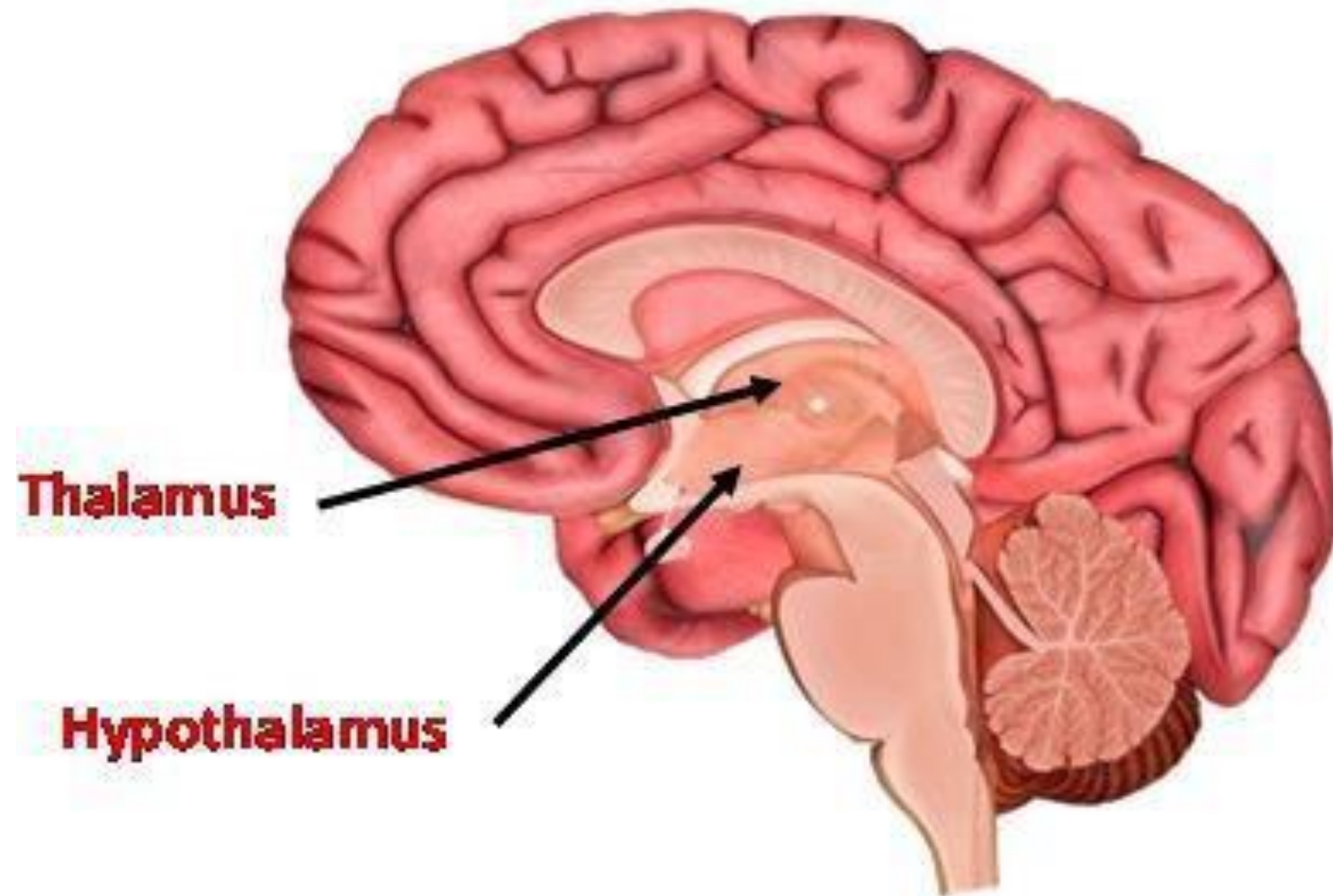


Cerebrum: formed of the 2 cerebral hemispheres each one is **FORMED OF 4 LOBES**) and the diencephalon.

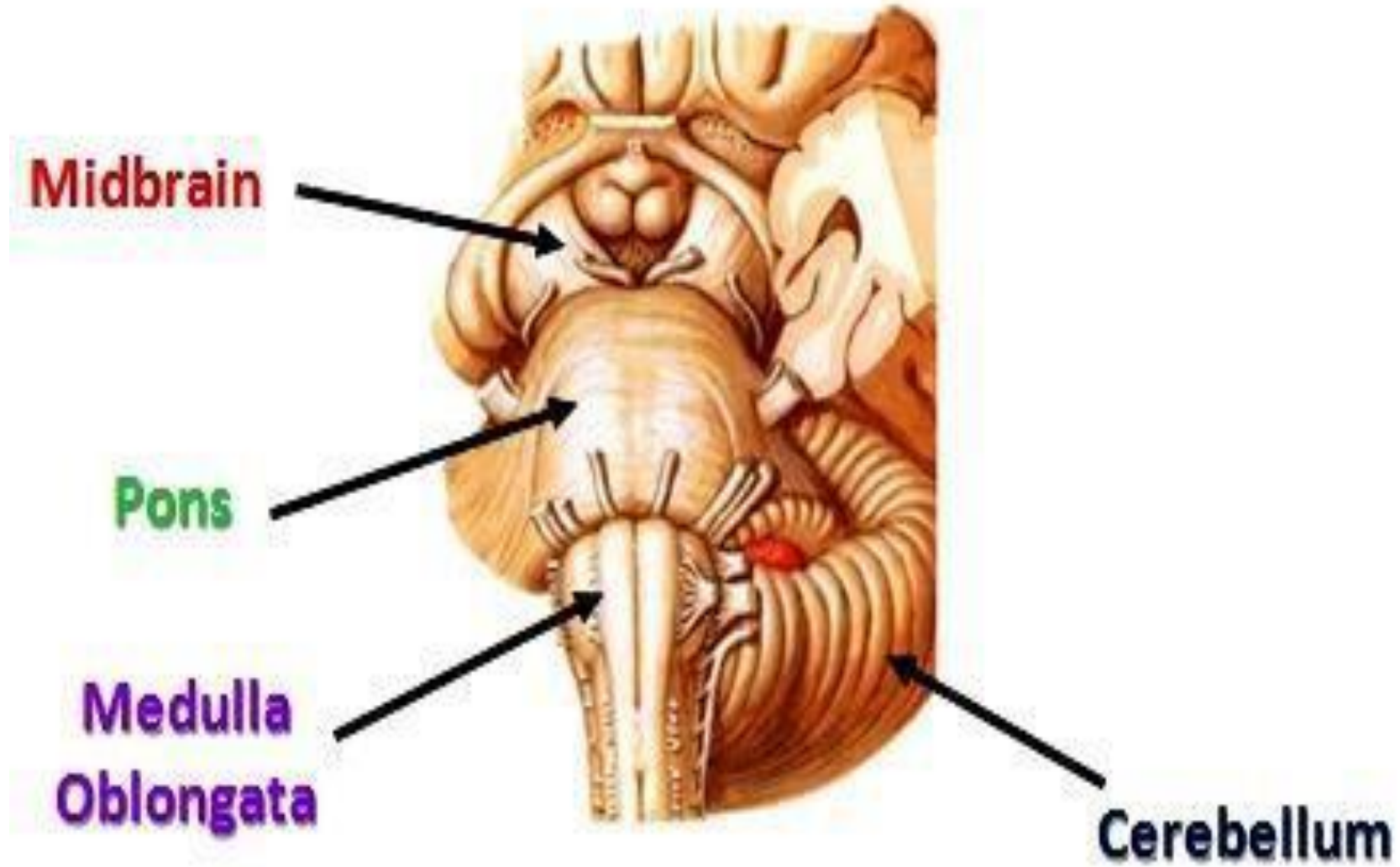


Lobes of the cerebral Hemisphere

Diencephalon



Brainstem & Cerebellum



Brain stem

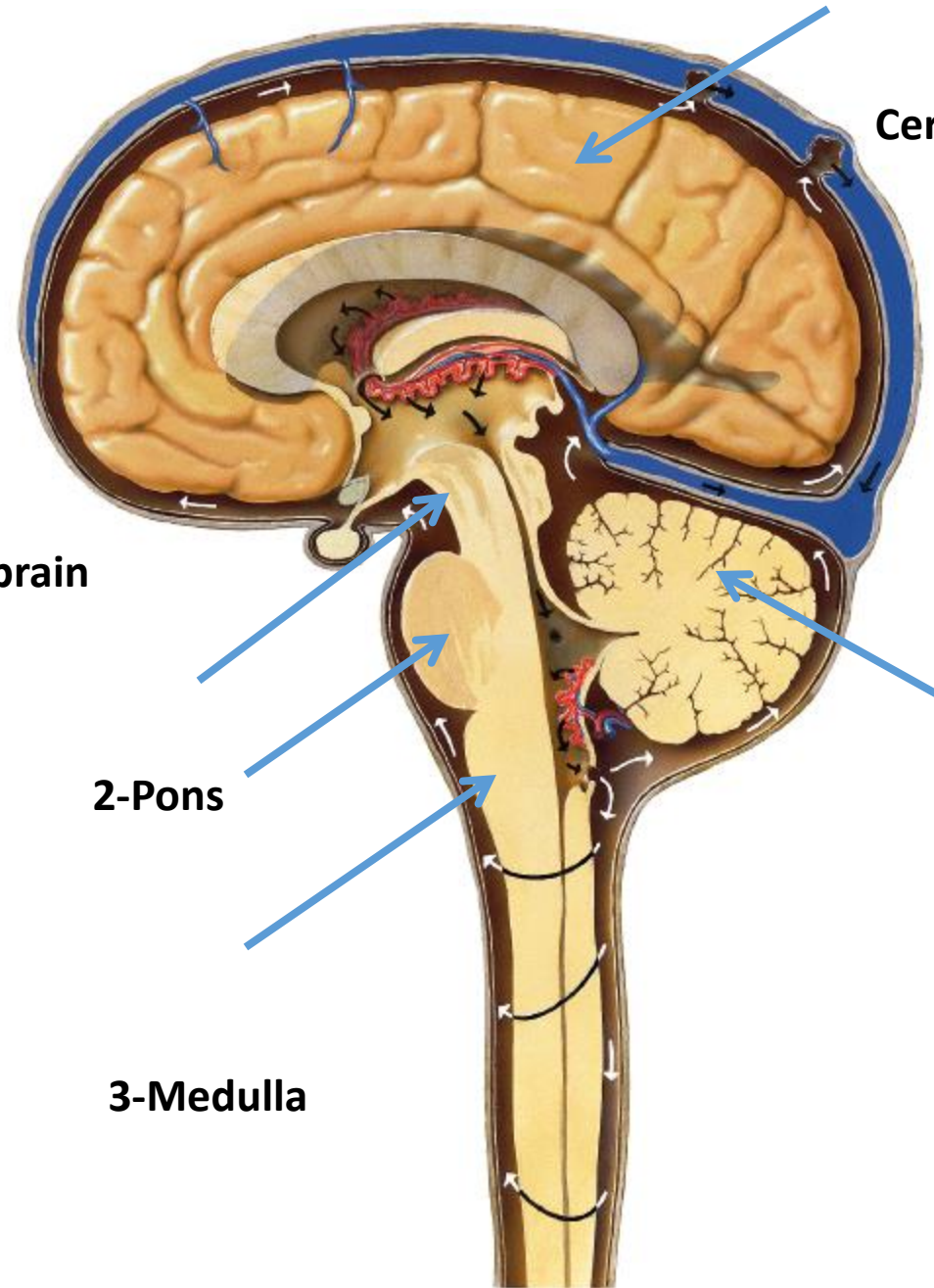
1-Mid brain

2-Pons

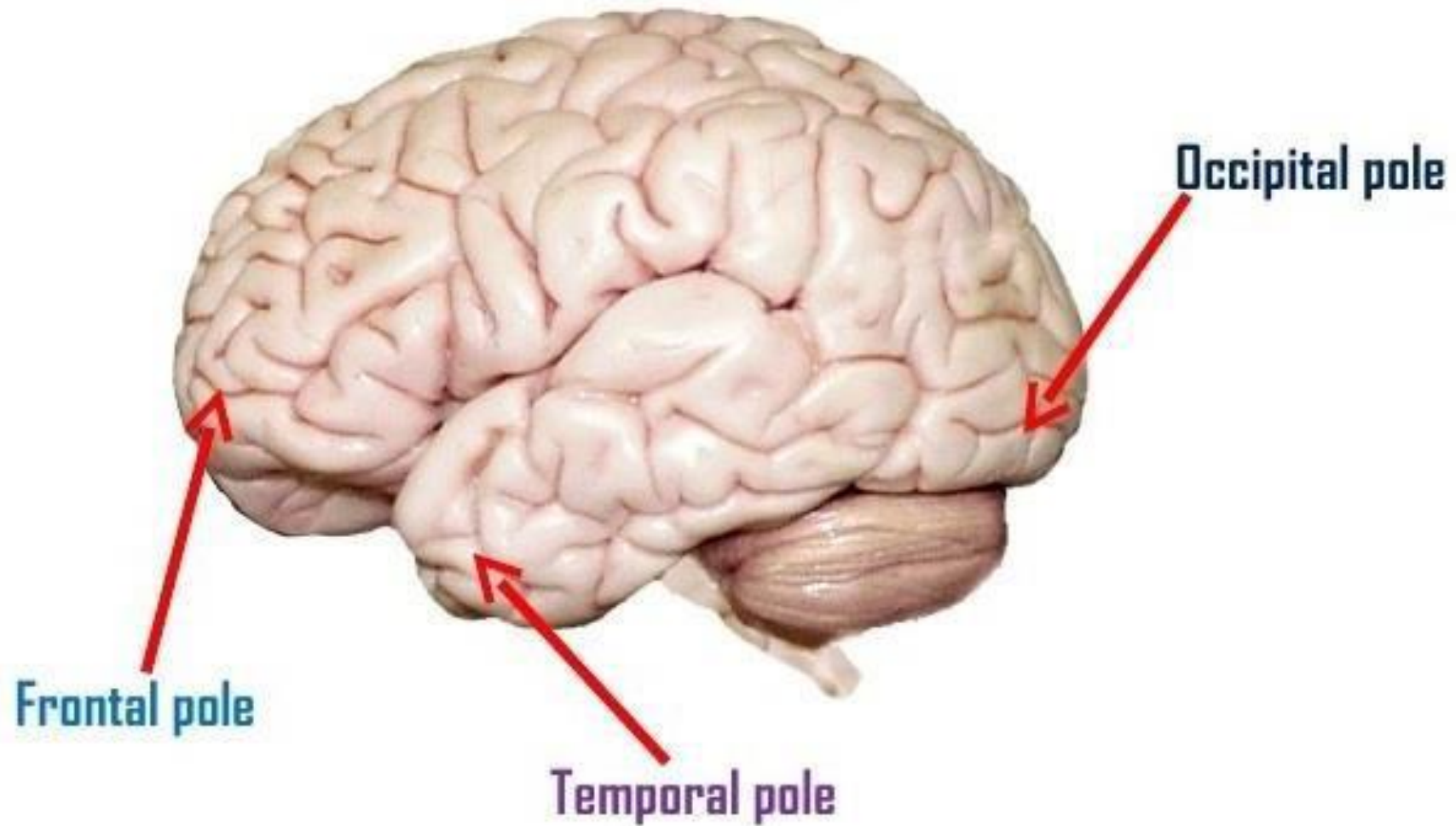
3-Medulla

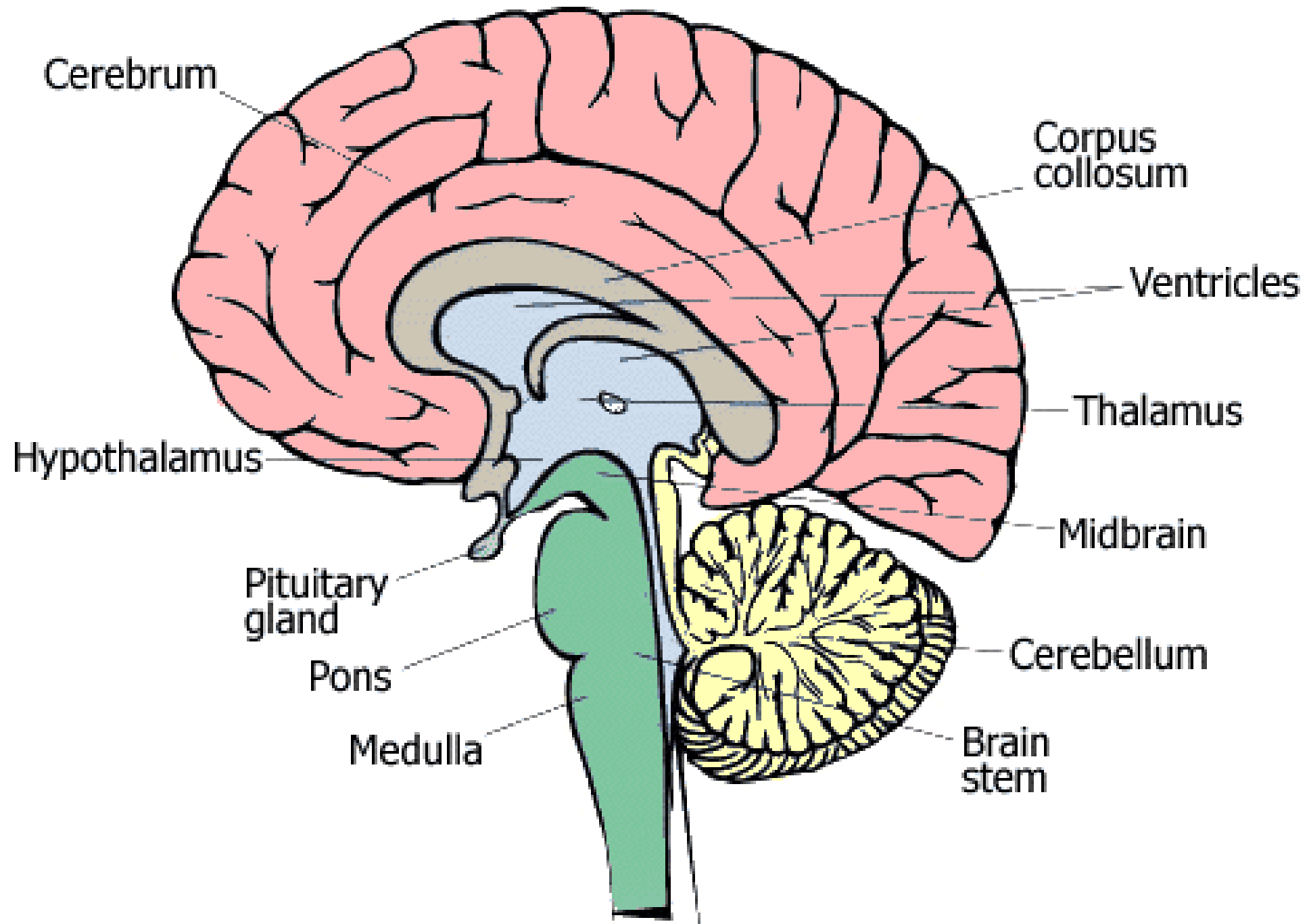
Cerebral hemisphere

cerebellum



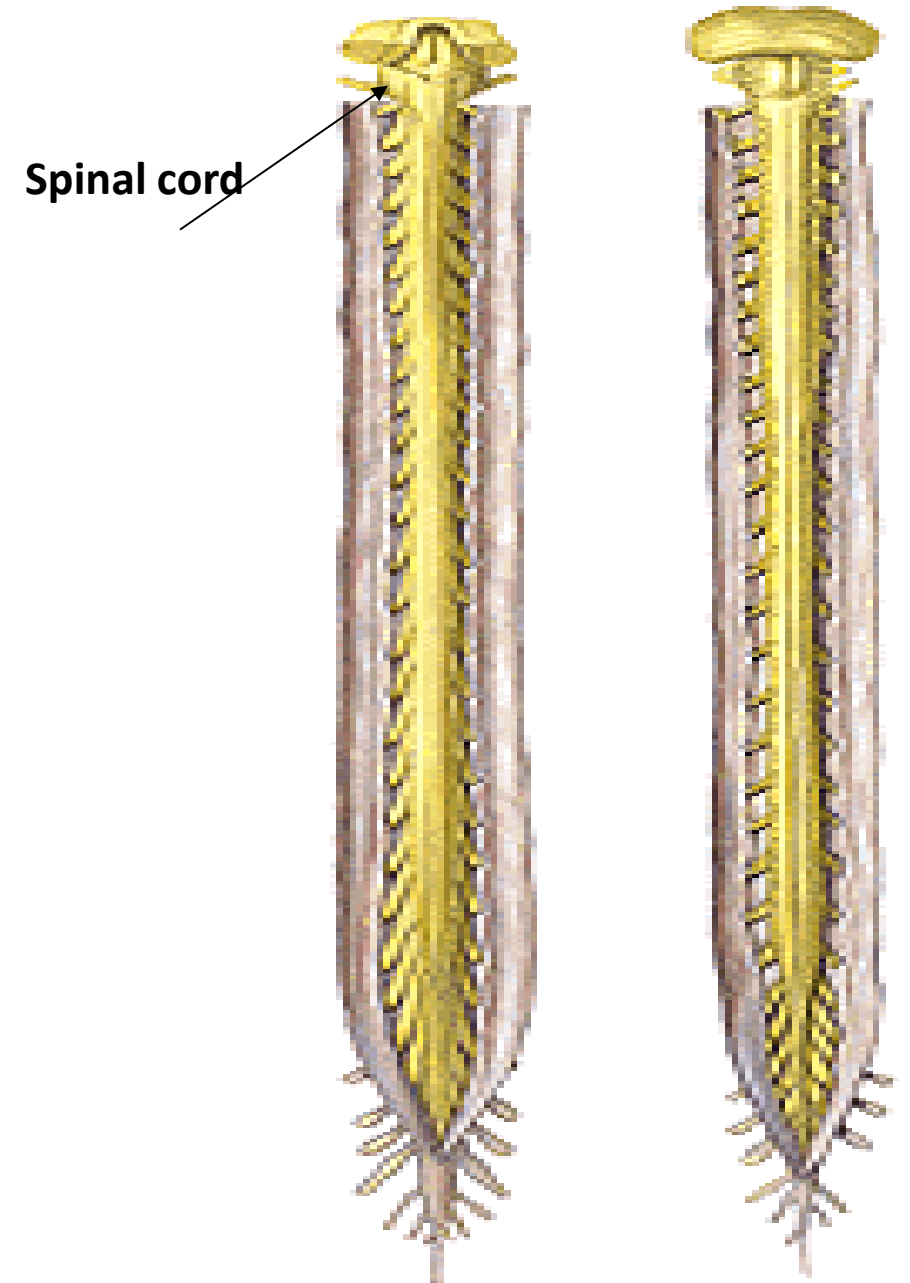
Poles of the cerebral Hemisphere





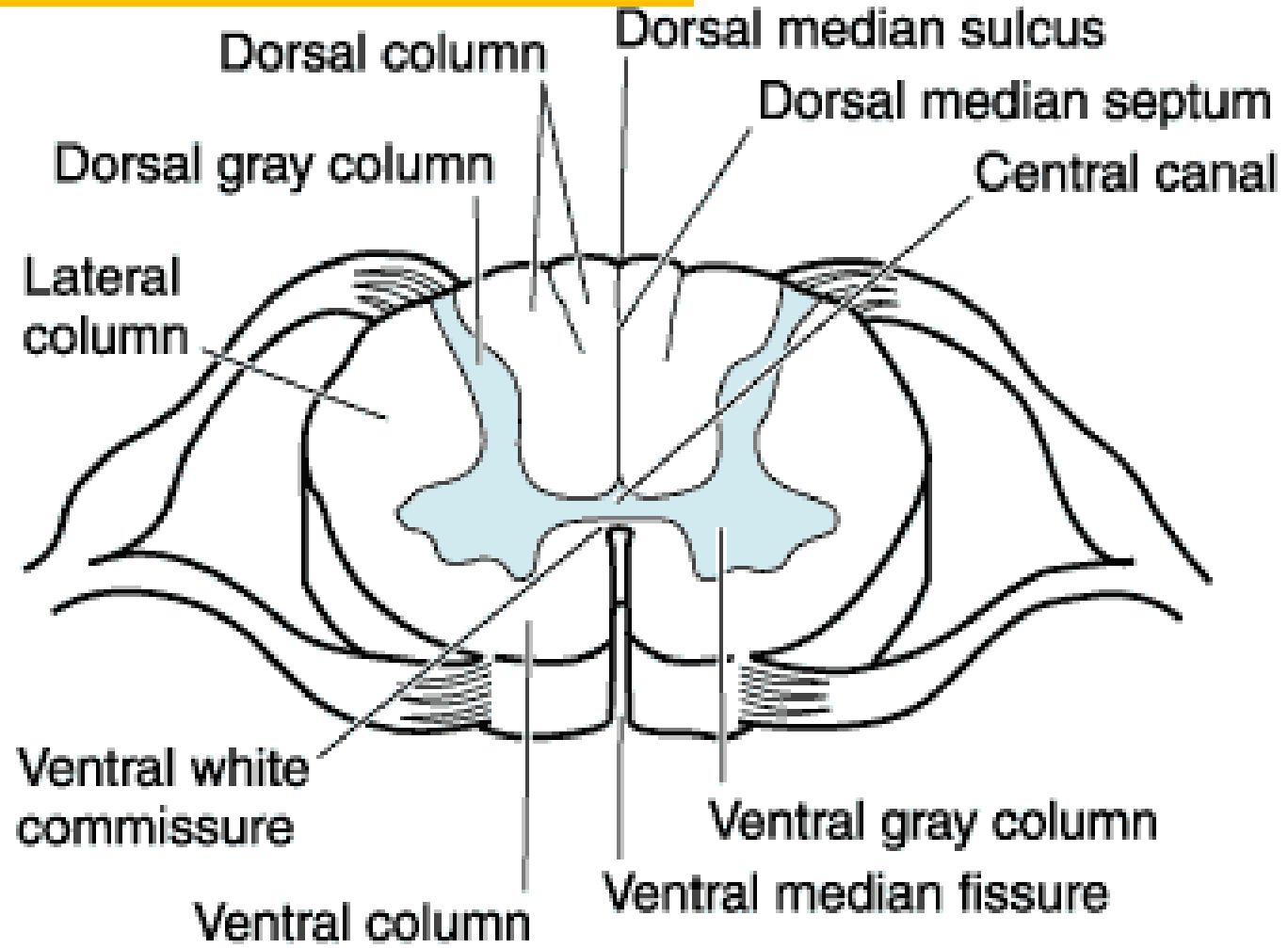
2-Spinal Cord

- Cylindrical
- 45 cms
- Direct continuation of **medulla oblongata**



Structure of Spinal Cord

- Inner core of **grey matter**
- Grey matter is H shaped **anterior horn** containing motor nuclei and **posterior horn** containing sensory nuclei
- Outer layer of **white matter** containing the fibers



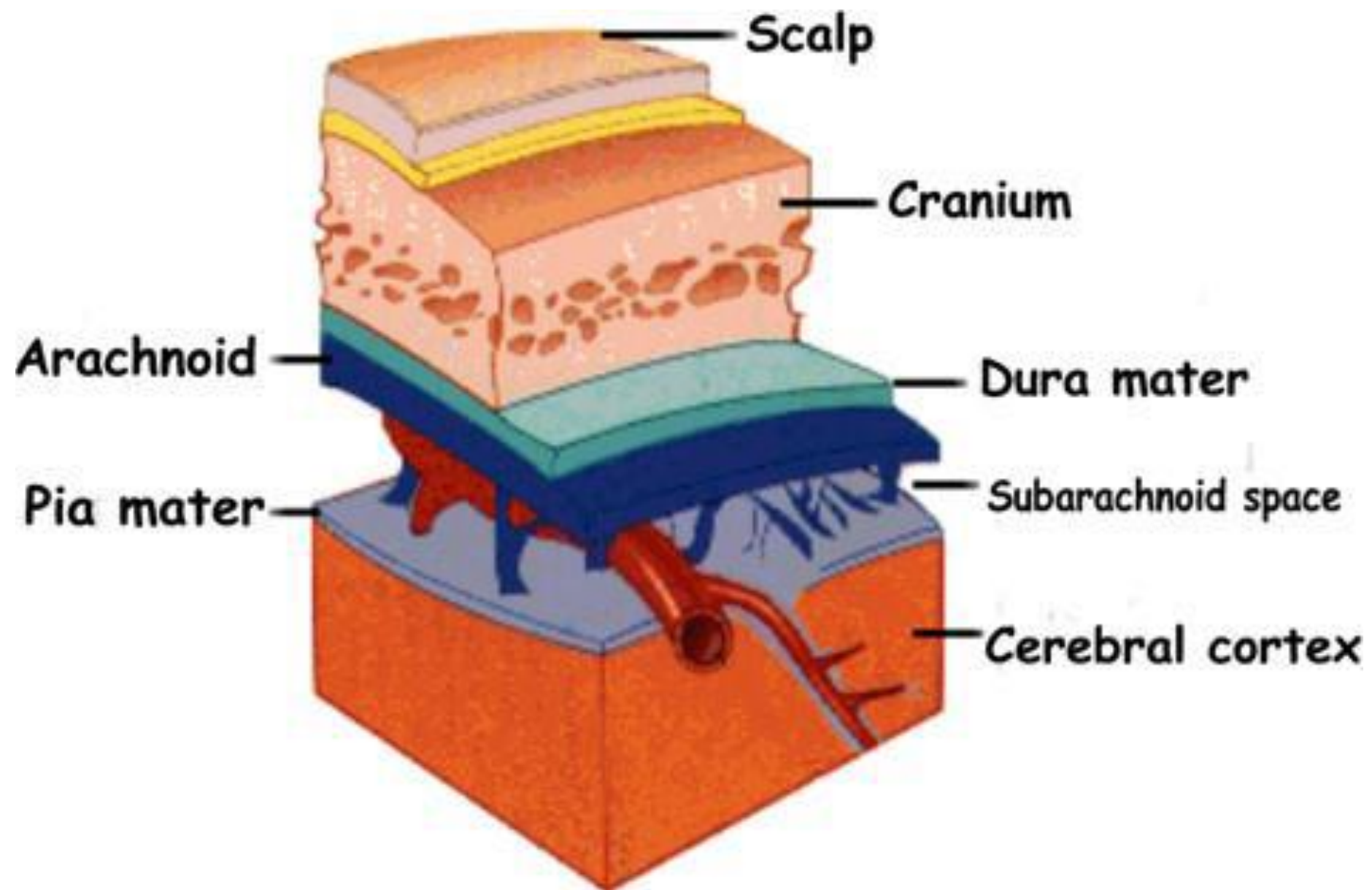
Coverings of brain and spinal cord=**Meninges**

1.Dura matter outermost

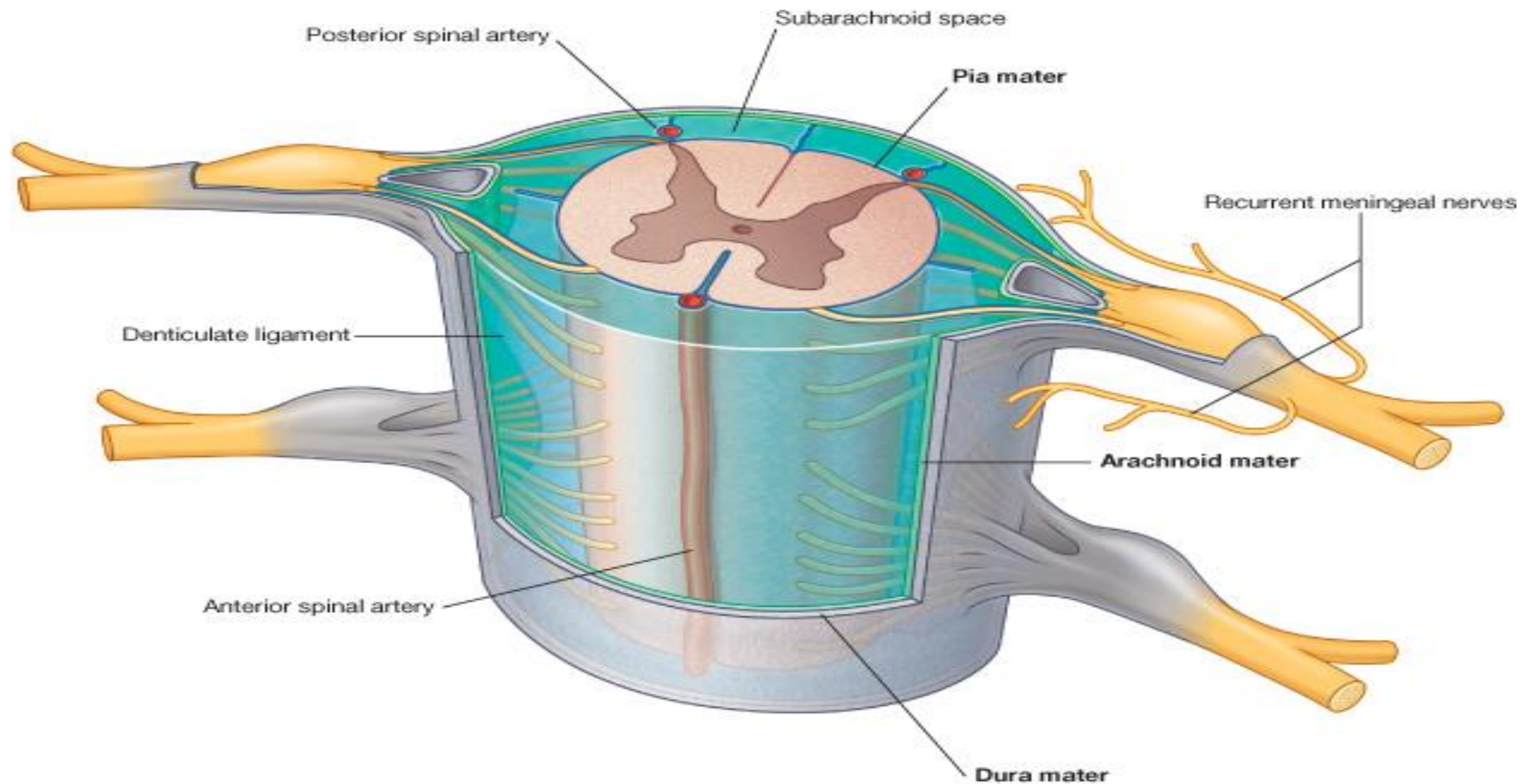
2.Arachnoid

3.Pia matter innermost

4.Three layers are called together **meninges**







2-Peripheral nervous system

2-The peripheral nervous system:

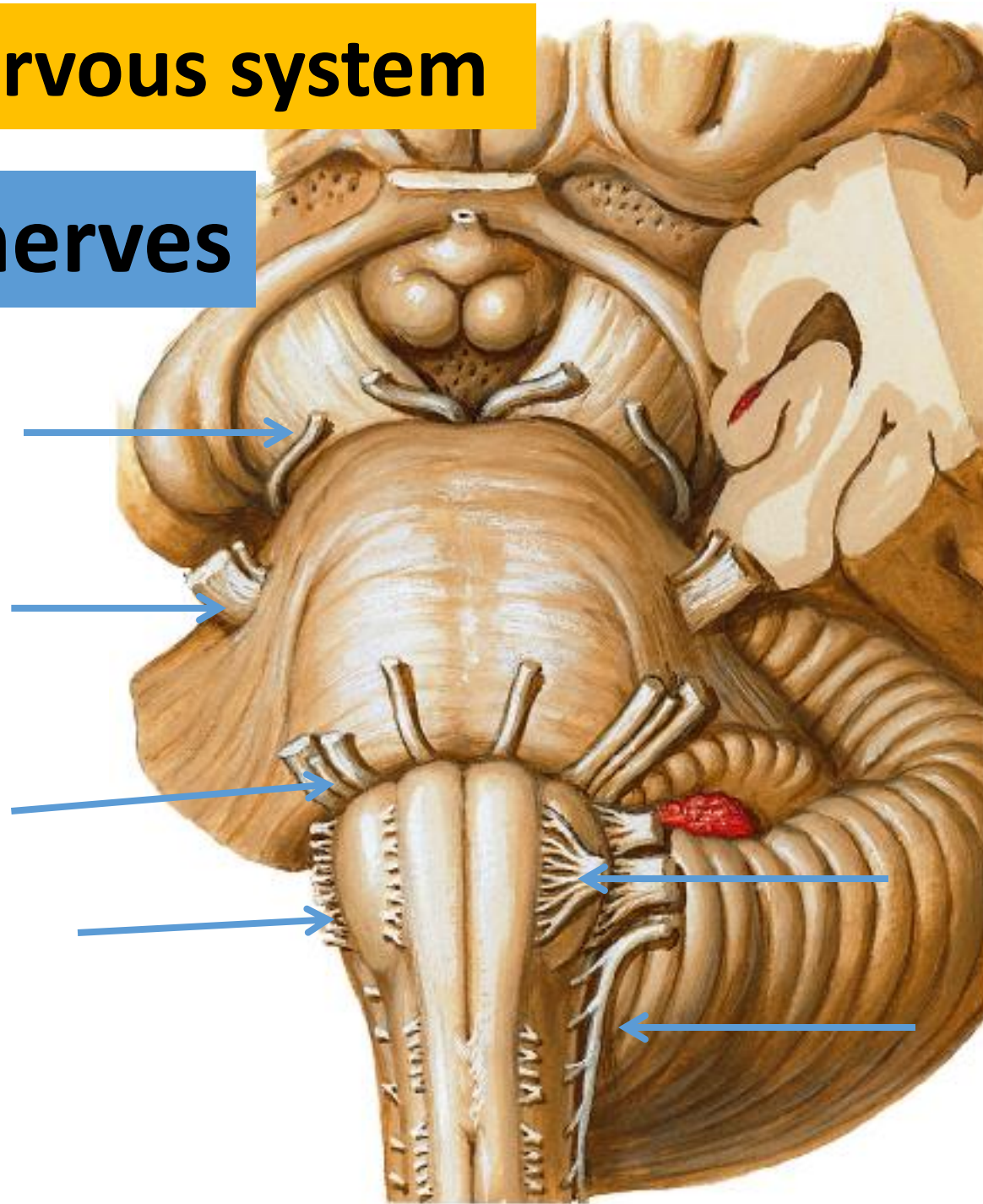
1) Cranial nerves :-

These are 12 pairs which are named as the following:

Number	Name	Number	Name
I	Olfactory	VII	Facial
II	Optic	VIII	Vestibulo-cochlear
III	Oculomotor	IX	Glossopharyngeal
IV	Trochlear	X	Vagus
V	Trigeminal	XI	Accessory
VI	Abducent	XII	Hypoglossal

Peripheral nervous system

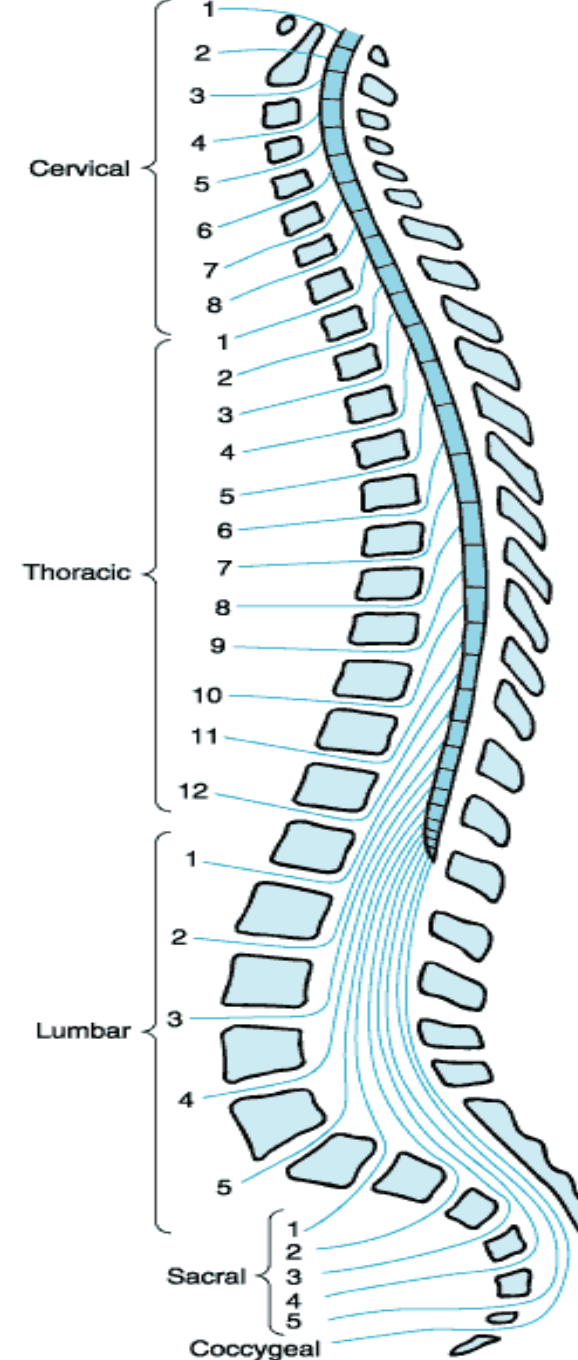
12 cranial nerves



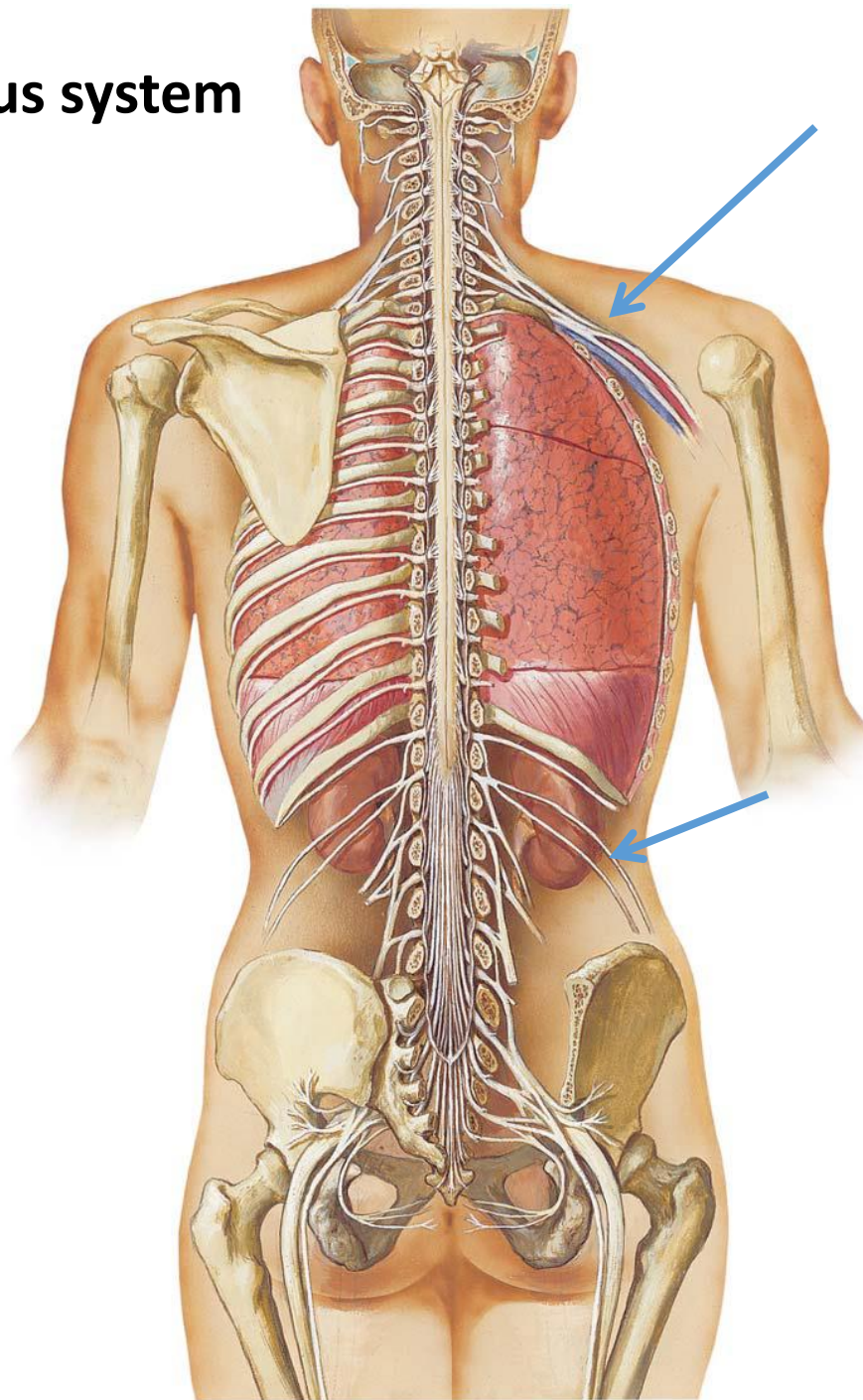
2-Spinal nerves

31 pairs of spinal nerves
are attached to spinal cord

- **8**, cervical
- **12**, thoracic
- **5**, lumbar
- **5**, sacral
- **1** coccygeal

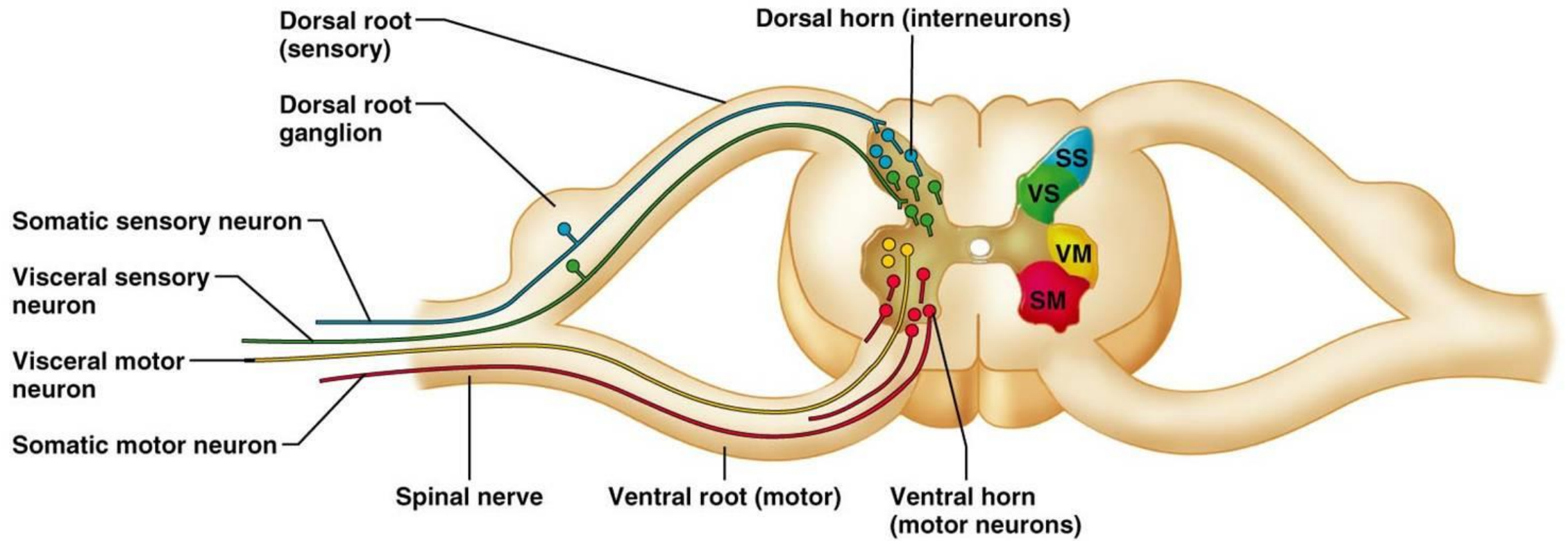


Peripheral nervous system
Spinal nerves 31



Spinal nerves

Spinal nerves



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Spinal Nerves

- **Dorsal & ventral roots** unit to form spinal nerve mixed TRUNK
- Spinal nerves divide to 2 rami
- **Ventral rami** : mixed for skin & muscles of **anterolateral** part of trunk & **tend to form plexus**
- **Dorsal rami** : mixed for skin & muscles of **back** of neck and back. **Don't form plexus**

